

Decision Log

Key Assumptions

- Monday.com is the main source of truth for the Deals/Pipeline and Work Orders data.
- The data may contain missing values, inconsistent dates, and different naming conventions.
- Users will ask business-level questions rather than questions about individual records.
- Historical data may not always be available, so comparisons are made only when sufficient data exists.

Trade-offs Chosen and Why

- MCP over direct API: Chosen because it provides a clean tool interface for the AI agent and makes it easier to add other data sources later.
- LLM + deterministic logic: The LLM handles query understanding and response generation, while Python handles calculations, filtering, and data cleaning for more reliable results.
- Single agent over multiple agents: Chosen to keep the system simpler and easier to test within the assignment scope.
- Backend processing over LLM calculations: Business metrics are calculated in the backend to reduce incorrect or inconsistent results.

What I Would Do Differently With More Time

- Add historical data snapshots for better week-over-week and month-over-month comparisons.
- Add root-cause analysis to answer questions such as “Why did the pipeline decrease?”
- Add scheduled leadership updates through email or Slack.
- Add source-level evidence so users can trace insights back to Monday.com records.
- Add anomaly detection for unusual changes in pipeline and operational metrics.

Interpretation of “Leadership Updates”

- I interpreted leadership updates as a short executive summary rather than a raw data report.
- The update would include key metrics, important changes, major wins, business risks, and areas requiring attention.
- The agent can combine information from both Deals/Pipeline and Work Orders to provide a broader business view.
- Data-quality issues that could affect the conclusions are also highlighted.
- The goal is to answer: What happened, why does it matter, and what should leadership focus on?